

Java-based Development of a Vehicle Speed Recommendation Model for Assisted Driving

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Abstract—With the rapid development of assisted driving technology, it has become especially critical to solve the problem of difficulty in determining the speed of assisted driving vehicles at intersections. In this paper, we propose a speed recommendation model based on multi-factor considerations, aiming to provide accurate speed recommendations for assisted driving vehicles to help them pass through intersections quickly and smoothly. The speed recommendation model integrates several key factors, including intersection type, traffic flow, and weather conditions. By combining and evaluating the effects of these factors, the model can determine the appropriate vehicle speed to ensure safe and efficient driving at the intersection. To verify the effectiveness of the model, we conducted simulation experiments using Java back-end development project and validated the model using simulated traffic data. The experimental results show that the speed suggestion model is able to provide accurate speed suggestions for assisted driving vehicles in different types of intersection scenarios, solving the problem of difficulty in determining vehicle speed at intersections. At the same time, the model has the potential to adapt and optimize, and can flexibly adjust the speed recommendations under different road conditions and traffic situations to meet the actual driving requirements. This has a positive impact on improving road safety and promoting the popularity of assisted driving technology.

Keywords—Assisted driving, Java system development, Multi-factor, Speed recommendation model

I. INTRODUCTION

With the rapid development of technology and the growing demand of society for safe driving, assisted driving systems have become an important research area in the modern automotive industry. Assisted driving systems use advanced sensors, artificial intelligence algorithms and communication technologies to provide drivers with diverse assistance functions to improve driving safety, driving efficiency and ride comfort. Research and applications in this area are important for reducing traffic accidents, optimizing traffic flow, and improving the driving experience. Among them, accurate prediction and control of vehicle speed plays a crucial role in ensuring smooth and safe traffic flow at intersections. In modern transportation systems, intersections are key locations where traffic flows converge and intersect. An efficient and safe intersection passage is critical to ensure the smoothness of the entire traffic system. However, the speed of an assisted-driving vehicle passing at an intersection faces a complex coupling of multiple factors, such as weather conditions, road conditions, and pedestrian behaviour[1][2][3]. Reasonable speed decisions can ensure smooth vehicle passage at intersections and avoid traffic accidents and congestions. Therefore, how to accurately predict vehicle speed and give a reasonable reference in the multi-factor

environment of intersections is one of the critical issues in the promotion of assisted driving technology.

Currently, many researchers have joined in the development of assisted driving systems. For example, Sethuraman R, Sellappan S et al. proposed a method based on AdaBoost multiclass support vector machine and cat-and-mouse optimizer for ADAS intrusion detection, which can accurately classify normal and abnormal activities of driving vehicles[4]; Liu H, Song X et al. proposed an assisted driving system based on lane-changing driving strategy. By using vehicle networking and autonomous driving technology, the system collects real-time data from the surrounding environment and uses a model predictive control algorithm based on rolling time range optimization to develop an optimal lane change strategy, which can reduce the loss time in the diversion zone by more than 60% in a mixed flow environment[5]; A C Ramachandra et al. proposed an advanced driver assistance based on lane detection, blind spot detection and pedestrian detection system with Canny edge detection and directional histogram methods for image processing, which can provide early warning and assistance functions during vehicle driving, effectively improving road safety and preventing accidents[6]. In addition, the development of assisted driving systems involves several strategies, such as adaptive cruise control and lane keeping assistance systems[7][8][9]. Researchers have proposed various intelligent algorithms and optimization methods for data processing and decision making to provide accurate driving recommendations and assisted operations[10][11].

Despite the significant progress in improving driving safety, assisted driving systems still face several technical and application challenges. For example, complex road environments, uncertain traffic conditions, and severe weather conditions[12][13]. To address the limitations of existing methods for assisted driving speed control, this thesis aims to address the problem of vehicle speed control in intersection passage and proposes a speed recommendation model based on multi-factor considerations. The model integrates several factors such as intersection type, traffic flow, and weather conditions to accurately predict the appropriate speed to ensure that the speed provided by the assisted driving system enables the vehicle to pass through the intersection safely and efficiently. The model employs machine learning and data analysis techniques that can learn and extract the relationship between vehicle speed and numerous factors from a large amount of simulated data to provide more accurate speed recommendations. In addition, a Java-based interface end has been developed in this thesis, and the Java back-end development project is used to conduct simulation experiments and validate the model with simulated traffic data. Through this interface, the assisted driving system can receive and process data from various sensors and perform speed

control based on the recommendations provided by the model to achieve a smooth passage through the intersection. The development and validation of the model also provides strong support and guidance for the further promotion and application of the assisted driving technology.

II. VEHICLE SPEED RECOMMENDATION BASE MODEL CONSTRUCTION

In this section, a base model is constructed based on the assisted driving system during a single traffic light phase cycle at the intersection. By defining the recommended vehicle speed ($V_{\text{suggested}}$) and the traffic congestion index (TI), a maximum recommended speed and a minimum recommended speed range are provided at 115 m before the intersection. Upon receiving the speed recommendation from the assisted driving system, the vehicle will slow down or accelerate accordingly to gradually adjust the speed to the recommended speed range. During the whole process, the driver's behaviour can be divided into three phases. First is the reaction process phase, where the driver receives the speed recommendation from the assisted driving system, reacts quickly, and starts to adjust the speed. Next is the acceleration or deceleration to the suggested speed phase, where the driver will gradually adjust the speed to within the suggested speed range. Finally, there is the uniform speed phase, once the driver reaches the recommended speed range, the vehicle will drive steadily at this speed until the intersection.

Based on the above, a base model was developed to describe an assisted driving system that provides a maximum recommended speed and a minimum recommended speed range at a position 115 meters before the intersection. The driver gradually adjusts the speed based on this recommendation to achieve a safe and smooth passage.

A. Recommended Speed Definition

Recommended speed $V_{\text{suggested}}$ Setting:

$$V_{R \min} \leq V_{\text{suggested}} \leq V_{G \max} \quad (1)$$

where, $V_{R \min}$ represents automatically drive the car without stopping to cross the line at the first phase of the green light during that traffic light cycle. $V_{G \max}$ represents that the car does not stop to cross the line at the end of the green phase or at the beginning of the red phase in this traffic light cycle.

The speed recommendation model uses a strategy based on random generation and reference value update to determine the current recommended speed $V_{\text{suggested}}$. This strategy uses a random number between $V_{R \min}$ and $V_{G \max}$ as the suggested speed and limits it to the reference value of the next generation range. If the current suggested speed does not guarantee safe passage through the intersection within the green phase, we use the next green start moment as the reference value for calculating $V_{\text{suggested}}$. This dynamic adjustment recommendation adapts to different traffic conditions and green light phase changes, providing drivers with flexible and effective speed recommendations to ensure safe vehicle passage.

B. Traffic Congestion Index Definition

Traffic congestion index (TI) on a specific section m:

$$TI_m = \frac{T_m}{T'_{m(\beta)}} = \frac{V'_{m(\beta)}}{V_m} \quad (2)$$

where $T'_{m(\beta)}$ denotes the time required for a vehicle to pass in a free state (i.e., when the traffic volume is 0) on road section m, T_m denotes the time required for the vehicle to actually pass on road section m, $V'_{m(\beta)}$ denotes the speed of the vehicle passing in the free state (i.e., when the traffic volume is 0) on roadway m and V_m denotes the actual speed of the vehicle passing on section m.

$T'_{m(\beta)}$ and $V'_{m(\beta)}$ reflects the state in the ideal traffic-free situation, while T_m and V_m reflects the effect of the actual traffic flow on the vehicle passing time and speed. Now making a definition of the relevant traffic flow for TI :

$$TI_m = \begin{cases} \text{Unblocked} & 0 \leq TI_m < 2 \\ \text{Basic Unblocked} & 2 \leq TI_m < 4 \\ \text{Light Congestion} & 4 \leq TI_m < 6 \\ \text{Moderate Congestion} & 6 \leq TI_m < 8 \\ \text{Heavy Congestion} & 8 \leq TI_m < 10 \end{cases}$$

C. Basic Model Construction

Assume measurement and model construction within 115m before a defined intersection. There are three phases within the specified distance: reaction phase, deceleration or acceleration to the recommended speed phase, and uniform speed phase.

1) Reaction phase

The driver needs to react quickly and decide on the appropriate speed after receiving a speed recommendation from the assisted driving system.

$$S_1 = V_{ini} \cdot T_{\text{react}} \quad (3)$$

where, V_{ini} denotes the initial speed of the vehicle during normal driving, and T_{react} denotes the reaction time for the driver to react after receiving the speed suggestion from the assisted driving system.

2) Deceleration or acceleration to the recommended speed phase

$$\begin{cases} S_2 = \frac{V_{\text{suggested}}^2 - V_{ini}^2}{2 \sum a_i} \\ T_{\text{act}} = \frac{|V_{\text{suggested}} - V_{ini}|}{\sum a_i} \end{cases} \quad (4)$$

3) Uniform speed phase

$$S_3 = V_{\text{suggested}} \cdot T_3 \quad (5)$$

$$V_{ini} \cdot T_{\text{react}} + \frac{|V_{\text{suggested}}^2 - V_{ini}^2|}{2 \cdot \sum a_i} + V_{\text{suggested}} \cdot T_3 = 115 \quad (6)$$

4) Vehicle speed recommendation model for the assisted driving system under the corresponding acceleration and deceleration strategies

a) Deceleration strategies

In the second stage above, the vehicle needs to complete a deceleration strategy based on the assisted driving system speed recommendation model.

$$V_{\text{suggested}} = \sqrt{2 \cdot \sum a_i \cdot (115 - V_{ini} \cdot T_{\text{react}}) + V_{ini}^2 + (\sum a_i \cdot T_3)^2} - \sum a_i \cdot T_3 \quad (7)$$

$$0 \leq T_3 = T_m - T_{react} - T_{action} \leq \frac{115}{V_{suggested}} \quad (8)$$

From equation (6)(7)(8), the proposed speed range under the acceleration strategy is obtained:

$$V_{suggested} \begin{cases} \geq V_{ini} - \sum a_i \cdot (T_m - T_{react}) \\ \leq \sqrt{115 \cdot \sum a_i + \frac{1}{4} \cdot [\sum a_i \cdot (T_m - T_{react}) - V_{ini}]^2} \\ - \frac{1}{2} \cdot \sum a_i \cdot (T_m - T_{react}) - \frac{1}{2} \cdot V_{ini} \end{cases} \quad (9)$$

b) Acceleration strategies

In the second stage above, the vehicle needs to complete an acceleration strategy based on the assisted driving system speed recommendation model.

$$V_{suggested} = \sqrt{V_{ini}^2 - 2 \cdot \sum a_i \cdot (115 - V_{ini} \cdot T_{react}) + (\sum a_i \cdot T_3)^2} + \sum a_i \cdot T_3 \quad (10)$$

$$\frac{115}{V_{suggested}} \leq T_3 = T_m - T_{react} - T_{action} \leq \frac{115}{V_{ini}} \quad (11)$$

From the three equations (6)(11)(12), the proposed speed range under the acceleration strategy is obtained:

$$V_{suggested} \begin{cases} \geq \sum a_i \cdot (T_m - T_{react}) - \frac{115 \cdot \sum a_i}{V_{ini}} - V_{ini} \\ \leq \sqrt{\frac{1}{4} \cdot [\sum a_i \cdot (T_m - T_{react}) - V_{ini}]^2 - 115 \cdot \sum a_i} \\ + \frac{1}{2} \cdot \sum a_i \cdot (T_m - T_{react}) + \frac{1}{2} \cdot V_{ini} \end{cases} \quad (12)$$

After the construction of the base model, the speed recommendation range provided by the speed recommendation model of the assisted driving system can be derived.

III. WEATHER-ADJUSTED SPEED RECOMMENDATION MODEL

Based on the above in-depth analysis of the base model, we considered the effect of weather conditions on the human body's reaction time to external stimuli and calculated the T_{react} in the base model. In this section, we take the three dimensions of temperature, humidity and wind speed as considerations and investigate their effects on the human body's reaction time after receiving external stimuli, and accordingly propose a suggested model for vehicle speed considering weather factors.

Based on the available studies, we conclude that, in general, humans usually need about 0.2 to 0.3 seconds to respond to a stimulus. Under suitable weather conditions, the minimum response time for adults is usually around 0.15 seconds. The appropriate temperature range is 17°C to 25°C, the appropriate humidity range is 40% to 60%, and the appropriate wind speed should be maintained between 0.15 and 0.25 meters per second.

By considering these weather factors, we can more accurately determine the speed recommendations for the assisted driving system. This helps provide more personalized and adaptive speed recommendations to ensure that drivers can react quickly and safely in different weather conditions and to facilitate smooth vehicle passage at intersections.

A. Data Simulation

1) Temperature

$$RT_t = RT_0 \cdot (1 + k_1 \cdot (T - T_0)) \quad (13)$$

where RT_t is the reaction time at temperature T , RT_0 is the reaction time at standard temperature T_0 , k_1 is a constant, and T is the current temperature.

When the temperature increases, the reaction time of the human body slows down. For every 1 °C increase in temperature, the human reaction time slows down by about 0.006 seconds, which is calculated as $k_1 = 0.04$.

2) Humidity

$$RT_h = RT_0 \cdot (1 + k_2 \cdot (H - H_0)) \quad (14)$$

where RT_h is the reaction time at humidity H , RT_0 is the reaction time at standard humidity H_0 , k_2 is a constant, and H is the current humidity.

When the humidity rises, the evaporative heat dissipation of the body slows down and the internal temperature of the body rises, resulting in a slower reaction time. According to research, when the relative humidity increases by 1%, the body's reaction time slows down by about 0.006 seconds, which gives $k_2 = 0.04$.

3) Wind speed

$$RT_v = RT_0 \cdot (1 + k_3 \cdot (V - V_0)^2) \quad (15)$$

where RT_v is the reaction time at wind speed V , RT_0 is the reaction time at the optimum wind speed (i.e., the shortest reaction time), V_0 is the optimum wind speed (about 0.2 m/s), k_3 is a constant, and V is the current wind speed.

When the wind speed increases, the heat dissipation from the body surface becomes faster, which leads to faster reaction time. According to the study, when the wind speed increases by about 0.002 seconds per 1 m/s, the human body's reaction time increases by about 0.002 s. As the wind speed increases, the muscular system requires more regulation and balance, resulting in the brain's attention and processing speed being distracted. From the linear fit and the root formula, we obtain the estimated value of $k_3 = 0.01064$.

4) Data pre-processing and cleaning

Characteristic scaling of temperature and wind speed: and calculation of their impact scores for temperature, humidity, and wind speed.

$$scaled_T_i = \frac{T_i - \min_T}{\max_T - \min_T} \quad (16)$$

$$scaled_V_{wi} = \frac{V_{wi} - \min_V_w}{\max_V_w - \min_V_w} \quad (17)$$

For the calculation of the temperature, humidity and wind speed impact score f :

$$\begin{cases} f_t = A \cdot (scaled_T_i - scaled_T_0)^2 \\ f_h = B \cdot (H - H_0)^3 \\ f_v = C \cdot (scaled_V_{wi})^2 \end{cases} \quad (18)$$

where A, B, C are adjustable parameters. $scaled_T_0$ is the scaled comfort temperature, H_0 is the standard humidity.

Using the matrix method, matrix operations are constructed for the three dimensions of temperature, humidity, and wind speed:

$$Y[:,0] = A \cdot (scaled_T_i - scaled_T_0)^2 + B \cdot (H - H_0)^3 + C \cdot scaled_V_{wi}^2 \quad (19)$$

From the random spanning tree $A + B + C = 100$, the effect of temperature humidity wind speed is obtained as a percentage of the response time for each group:

$$RT = RT_s + \frac{f_t}{Y[:,0]} \cdot (RT_t - RT_0) + \frac{f_h}{Y[:,0]} \cdot (RT_h - RT_0) + \frac{f_v}{Y[:,0]} \cdot (RT_v - RT_0) \quad (20)$$

Through the above algorithm, 100000 sets of data were obtained and the values with impact scores less than 0.1 and greater than 0.9 were excluded to obtain the simulated data shown in Figures 1 and 2.

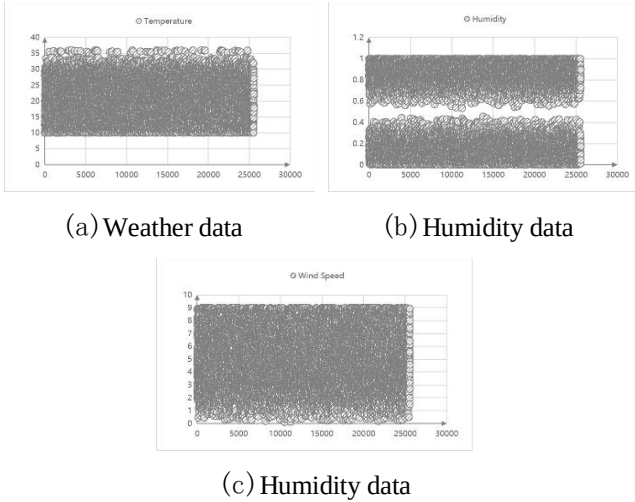


Fig. 1. Simulated data for three weather factors



Fig. 2. Reaction time distribution set

As shown in the Figure 2, the simulated data algorithm data is feasible and yields a response time (RT) distribution between 0.24 and 0.255, which is consistent with the judgment made based on existing studies.

B. Model Construction of Assisted Driving Speed Recommendation Based on Weather Factors

Based on the simulation data obtained above, a new weather model is constructed, and the algorithm incorporates some knowledge of meteorology. The model is constructed with reference to the PMV/PPD model and is applicable to a certain range of environments and conditions.

1) Algorithm construction

The body sensory temperature (AT) is calculated based on temperature, humidity and wind speed, and the algorithm accurately reflects the indicators of the human body's perception of the external environment. The calculation of body temperature index AT :

$$AT_i = 37 - \frac{37 - T_i}{0.68 - 0.0014 \cdot R_H + \frac{1}{1.76 + 1.4 \cdot V_i^{0.75}}} \quad (21)$$

$$R_H = \frac{H_i}{H_0} \quad (22)$$

where R_H is relative humidity (%) and V is wind speed (m/s).

Wind chill temperature (WT) is calculated based on wind speed, which is an index reflecting the change in external wind speed felt by the human body. Calculate the wind chill temperature index WT :

$$WT_i = 13.12 + 0.6215 \cdot T_i - 11.37 \cdot V_i^{0.16} + 0.39 \cdot T_i \cdot V_i^{0.16} \quad (23)$$

where, T_i denotes the actual temperature, V_i denotes the wind speed.

The relationship between the response time and the two indices is obtained by linear fitting of the two obtained indices:

$$W_{RT} = k_1 \cdot AT_i + k_2 \cdot WT_i \quad (24)$$

Then,

$$W_{RT} = k_1 \cdot \left(37 - \frac{37 - T_i}{0.68 - 0.0014 \cdot R_H + \frac{1}{1.76 + 1.4 \cdot V_i^{0.75}}} \right) + k_2 \cdot (13.12 + 0.6215 \cdot T_i - 11.37 \cdot V_i^{0.16} + 0.39 \cdot T_i \cdot V_i^{0.16}) \quad (25)$$

2) Model training

The dataset is first divided into a training set, a validation set, and a test set. The adopted ratio is 70% for training, to learn the patterns and relationships of the data through model training; 15% for validation to select hyperparameters of the model and perform model tuning; and 15% for testing to evaluate the performance of the model.

Then the model is initialized, and loss functions and optimizers are defined. By iterating, the features of the training set are fed into the model, and the parameters of the model are updated, and the value of the loss function is reduced according to the difference between the model prediction results and the actual labels. The procedure is

repeated until the model converges or reaches a predetermined number of training rounds. The model training yields:

$$k1 = -0.038989750152861276$$

$$k2 = 0.044860477297004056$$

Finally, the trained model is evaluated using the test set. For this model, the following evaluation values were obtained, as shown in Table 1.

TABLE I. EVALUATION VALUES FOR MODEL BASED ON WEATHER FACTORS

MSE	0.0001516771925512655
R-squared	0.7930647951477887
MAE	0.009772593515208436
RMSE	0.012315729477025123
MRE	0.03899540657194827%
Explained Variance	0.793102233200152

Based on the above evaluation data, it can be concluded that the weather factor-based speed recommendation model performs well in predicting vehicle speed with a certain degree of accuracy and reliability.

3) Aided driving model based on weather factors

a) For deceleration strategies:

$V_{suggested}$

$$\left\{ \begin{aligned} &\geq V_{ini} - \sum a_i \cdot \left(T_m - \left(k_1 \cdot \frac{37 - T_i}{0.68 - 0.0014 \cdot R_H + \frac{1}{1.76 + 1.4 \cdot V_i^{0.75}}} + k_2 \cdot \left(\frac{13.12 + 0.6215 \cdot T_i - 11.37 \cdot V_i^{0.16} + 0.39 \cdot T_i \cdot V_i^{0.16}}{11.37 \cdot V_i^{0.16} + 0.39 \cdot T_i \cdot V_i^{0.16}} \right) \right) \right) \\ &\leq \left[\sum a_i \cdot \left(T_m - \left(k_1 \cdot \frac{37 - T_i}{0.68 - 0.0014 \cdot R_H + \frac{1}{1.76 + 1.4 \cdot V_i^{0.75}}} + k_2 \cdot \left(\frac{13.12 + 0.6215 \cdot T_i - 11.37 \cdot V_i^{0.16} + 0.39 \cdot T_i \cdot V_i^{0.16}}{11.37 \cdot V_i^{0.16} + 0.39 \cdot T_i \cdot V_i^{0.16}} \right) \right) \right) \right]^2 \\ &\quad - 115 \cdot \sum a_i + \frac{1}{4} \cdot \left[\sum a_i \cdot \left(T_m - \left(k_1 \cdot \frac{37 - T_i}{0.68 - 0.0014 \cdot R_H + \frac{1}{1.76 + 1.4 \cdot V_i^{0.75}}} + k_2 \cdot \left(\frac{13.12 + 0.6215 \cdot T_i - 11.37 \cdot V_i^{0.16} + 0.39 \cdot T_i \cdot V_i^{0.16}}{11.37 \cdot V_i^{0.16} + 0.39 \cdot T_i \cdot V_i^{0.16}} \right) \right) \right) \right]^2 \\ &\quad - \frac{1}{2} \cdot \sum a_i \cdot (T_m - T_{react}) - \frac{1}{2} \cdot V_{ini} \end{aligned} \right. \quad (26)$$

b) For acceleration strategies:

$V_{suggested}$

$$\left\{ \begin{aligned} &\geq \sum a_i \cdot \left(T_m - \left(k_1 \cdot \frac{37 - T_i}{0.68 - 0.0014 \cdot R_H + \frac{1}{1.76 + 1.4 \cdot V_i^{0.75}}} + k_2 \cdot \left(\frac{13.12 + 0.6215 \cdot T_i - 11.37 \cdot V_i^{0.16} + 0.39 \cdot T_i \cdot V_i^{0.16}}{11.37 \cdot V_i^{0.16} + 0.39 \cdot T_i \cdot V_i^{0.16}} \right) \right) \right) \\ &\quad - \frac{115 \cdot \sum a_i}{V_{ini}} - V_{ini} \\ &\leq \left[\sum a_i \cdot \left(T_m - \left(k_1 \cdot \frac{37 - T_i}{0.68 - 0.0014 \cdot R_H + \frac{1}{1.76 + 1.4 \cdot V_i^{0.75}}} + k_2 \cdot \left(\frac{13.12 + 0.6215 \cdot T_i - 11.37 \cdot V_i^{0.16} + 0.39 \cdot T_i \cdot V_i^{0.16}}{11.37 \cdot V_i^{0.16} + 0.39 \cdot T_i \cdot V_i^{0.16}} \right) \right) \right) \right]^2 \\ &\quad - 115 \cdot \sum a_i \\ &\quad + \frac{1}{2} \cdot \sum a_i \cdot \left(T_m - \left(k_1 \cdot \frac{37 - T_i}{0.68 - 0.0014 \cdot R_H + \frac{1}{1.76 + 1.4 \cdot V_i^{0.75}}} + k_2 \cdot \left(\frac{13.12 + 0.6215 \cdot T_i - 11.37 \cdot V_i^{0.16} + 0.39 \cdot T_i \cdot V_i^{0.16}}{11.37 \cdot V_i^{0.16} + 0.39 \cdot T_i \cdot V_i^{0.16}} \right) \right) \right) \\ &\quad + \frac{1}{2} \cdot V_{ini} \end{aligned} \right. \quad (27)$$

Based on the influence of human body's response to external weather factors, this study can quantify the influence of weather factors on driver's response time through comprehensive analysis of three factors and optimize a weather factor-based assisted driving speed recommendation model based on the basic model in Section 2 to improve driving safety and comfort.

IV. SPEED RECOMMENDATION MODEL CONSIDERING TRAFFIC FACTORS

In this section of the study, we use Tessng software to model and simulate the intersection sections and construct them in detail for different number of lanes and vehicle behaviours. Specifically, we constructed simulated lane models with lane numbers of 3, 4, 5, and 6, and considered different vehicle behaviours, including three vehicle directions (one right turn, one straight ahead, one left turn), four vehicle directions (one right turn, two straight ahead, one left turn), five vehicle directions (two right turns, two straight

ahead, one left turn), and six vehicle directions (two right turns, two straight ahead, two left turns). In the model, we also consider different vehicle types, including small cars, trucks, buses and other types of vehicles. Through extensive simulation experiments, we obtained detailed data on the average queue length at the traffic light intersection, the number of vehicles in the queue, and the travel time and distance traveled in the segment before the traffic light intersection.

A. Traffic Modelling Data Simulation and Traffic Factor Algorithm Construction

When the number of lanes is 3, the traffic simulation data is shown in the Figure 3:



Fig. 3. Data on travel distance and time, number of vehicles, and traffic flow when the number of lanes is 3.

When the number of lanes is 4, the traffic simulation data is shown in the Figure 4:



Fig. 4. Data on travel distance and time, number of vehicles, and traffic flow when the number of lanes is 4.

When the number of lanes is 5, the traffic simulation data is shown in the Figure 5:



Fig. 5. Data on travel distance and time, number of vehicles, and traffic flow when the number of lanes is 5.

When the number of lanes is 6, the traffic simulation data is shown in the Figure 6:



Fig. 6. Data on travel distance and time, number of vehicles, and traffic flow when the number of lanes is 6.

Based on the large amount of data obtained, a new algorithm is proposed, aiming to quantify the traffic congestion and calculate the traffic congestion index TI. This

algorithm integrates various data for different vehicle types and number of lanes, providing an important reference for traffic management and decision making. the algorithm from the above data is constructed:

$$T = T_0 + \left(\frac{S_i}{V_i} \right) \cdot \left(1 + \frac{C \cdot N}{V_i \cdot W_i} \right) \quad (28)$$

where T is the total time required for a vehicle to pass this section of road before the traffic light intersection (s), T_0 is the time required for a vehicle to pass this section of road without traffic congestion (s), S is the length of a vehicle passing this section of road (m), V is the average speed of a vehicle passing this section of road (m/s), N is the flow of a vehicle passing this section of road (vehicle/s), C is a factor indicating the interaction effect, and W is the width (m) of different types of vehicles.

Defining the number of lanes, vehicle direction, and vehicle type weights, the traffic congestion index is obtained as follows:

$$\begin{cases} Lane_weights = \{3:1, 4:0.8, 5:0.6, 6:0.4\} \\ Direction_weights = \begin{cases} 'Single\ Rightward': 0.5, \\ 'Single\ Straightward': 1, \\ 'Single\ Leftward': 0.8, \\ 'Two\ Rightward': 0.3, \\ 'Two\ Straightward': 0.6, \\ 'Two\ Leftward': 0.4 \end{cases} \\ Vehicle_type_weights = \begin{cases} 'Small\ cars': 1, 'Trucks': 0.8, \\ 'Buses': 0.6, 'Others': 0.4 \end{cases} \end{cases}$$

Then, about TI's algorithm construction is complete:

$$TI = L_w \cdot D_w \cdot N \quad (29)$$

$$TI = L_w \cdot D_w \cdot \left[\frac{V_i}{S_i} \cdot (T - T_0) - 1 \right] \cdot \frac{V_i \cdot V_w \cdot W}{C} \quad (30)$$

where, L_w represents the influence weight of the number of lanes, D_w represents the influence weight of vehicle direction, and V_w influence weight of vehicle type.

$$TI = \begin{cases} L_{w[3]} \cdot \frac{D_{w[SR]} + D_{w[SS]} + D_{w[SL]}}{\sum D_w} \cdot \left[\frac{V_i}{S_i} \cdot (T - T_0) - 1 \right] \cdot \frac{V_i \cdot V_w \cdot W}{C} & lane=3 \\ L_{w[4]} \cdot \frac{D_{w[SR]} + D_{w[TS]} + D_{w[SL]}}{\sum D_w} \cdot \left[\frac{V_i}{S_i} \cdot (T - T_0) - 1 \right] \cdot \frac{V_i \cdot V_w \cdot W}{C} & lane=4 \\ L_{w[5]} \cdot \frac{D_{w[TR]} + D_{w[TS]} + D_{w[SL]}}{\sum D_w} \cdot \left[\frac{V_i}{S_i} \cdot (T - T_0) - 1 \right] \cdot \frac{V_i \cdot V_w \cdot W}{C} & lane=5 \\ L_{w[6]} \cdot \frac{D_{w[TR]} + D_{w[TS]} + D_{w[TL]}}{\sum D_w} \cdot \left[\frac{V_i}{S_i} \cdot (T - T_0) - 1 \right] \cdot \frac{V_i \cdot V_w \cdot W}{C} & lane=6 \end{cases} \quad (31)$$

The training model is substituted to obtain $C = -0.1018326$.

According to this model, the mean square error obtained is 0.0000417482935548328, which indicates that the speed suggestion model considering traffic factors performs well in predicting vehicle speeds with some accuracy and reliability.

The model predicts the speed by considering traffic factors such as number of lanes, direction of traffic, vehicle type and traffic flow. The way these factors affect the speed is incorporated into the model, thus improving the accuracy of the prediction. The advantage of considering traffic factors on speed suggestion models is that they can consider real road conditions more comprehensively, rather than relying only on the characteristics of the vehicles themselves. The number of lanes is considered to reflect the width and capacity of the road, while the weight assignment of vehicle directions can reflect the impact of different turns on traffic flow. In addition, the weights of different vehicle types can consider the speed and handling characteristics of different vehicles and the differences in the space they occupy on the road.

B. Aided Driving Model Based on Traffic Factors

Based on the obtained traffic congestion index, we construct a new speed recommendation model. In the acceleration strategy, we provide the corresponding speed recommendation based on the traffic congestion index TI calculated by the algorithm. Similarly, for the deceleration strategy, we provide appropriate speed recommendations based on the traffic congestion index TI .

$$T_m = TI_m \cdot T'_{m(\beta)} \quad (32)$$

A new congestion index-based speed recommendation model can be constructed from the above equation:

1) Acceleration strategy:

$$V_{suggested} \begin{cases} \geq V_{ini} - \sum a_i \cdot (TI_m \cdot T'_{m(\beta)} - T_{react}) \\ \leq \sqrt{115 \cdot \sum a_i + \frac{1}{4} \cdot [\sum a_i \cdot (TI_m \cdot T'_{m(\beta)} - T_{react}) - V_{ini}]^2} \\ - \frac{1}{2} \cdot \sum a_i \cdot (TI_m \cdot T'_{m(\beta)} - T_{react}) - \frac{1}{2} \cdot V_{ini} \end{cases} \quad (33)$$

2) Deceleration strategy:

$$V_{suggested} \begin{cases} \geq \sum a_i \cdot (TI_m \cdot T'_{m(\beta)} - T_{react}) - \frac{115 \cdot \sum a_i}{V_{ini}} - V_{ini} \\ \leq \sqrt{\frac{1}{4} \cdot [\sum a_i \cdot (TI_m \cdot T'_{m(\beta)} - T_{react}) - V_{ini}]^2 - 115 \cdot \sum a_i} \\ + \frac{1}{2} \cdot \sum a_i \cdot (TI_m \cdot T'_{m(\beta)} - T_{react}) + \frac{1}{2} \cdot V_{ini} \end{cases} \quad (34)$$

By using the model, more accurate speed recommendations can be provided to help drivers better plan their routes and predict their arrival times. This can have a positive impact on reducing traffic congestion, improving road efficiency, and enhancing traffic safety. However, it should be noted that this model is only an example and may require further optimization and validation in real-world applications.

V. JAVA SIMULATION PROJECT CONSTRUCTION

A. Engineering Model

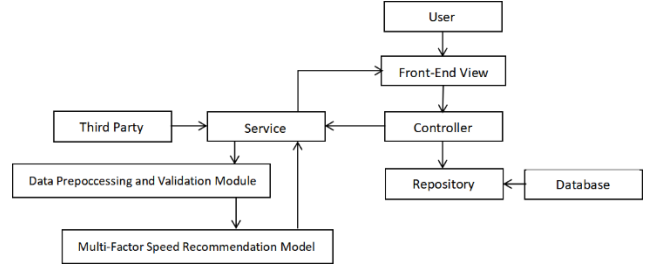


Fig. 7. Java back-end flowchart

According to Figure 7, this Java back-end simulation project uses the MVC model, in which the service layer manages the business logic, invokes the speed recommendation model with multi-factor considerations and the data pre-processing and validation module, and interacts with third-party modules to complete the user and controller delivery. The data pre-processing and validation module pre-processes and validates user input related data such as temperature, humidity and wind speed data to ensure the integrity of the data. The third-party module is responsible for communicating with third-party modules such as temperature, humidity and wind speed sensors, speed detectors, etc., and obtaining the required data from them when needed. The controller is responsible for receiving requests from the front-end and passing the request parameters to the service layer for processing. The repository management interacts with the database to store and pass the relevant data obtained from the controller to the back-end database for easy retrieval of data related to speed recommendations. The multi-factor speed recommendation model is the previous model which integrates factors such as traffic flow, and weather conditions to calculate the appropriate speed, Figure 8 below shows the Java back-end simulation project use case diagram.

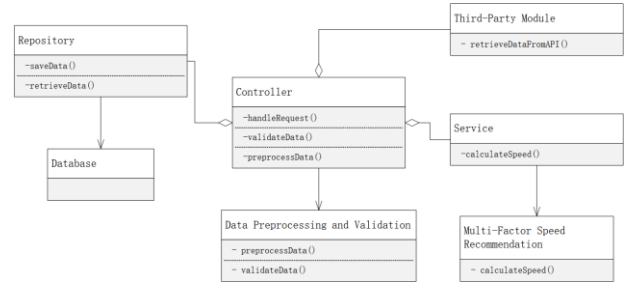


Fig. 8. Back-end implementation use case diagram

B. Introduction to User Visualization Interface

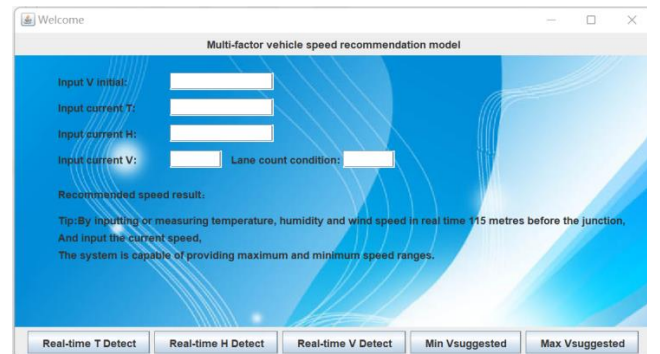


Fig. 9. Java front-end visual interface system

The above Figure 9 shows the front-end interface of the Java simulation project for the speed recommendation model of the assisted driving system. By inputting in real time or measuring in real time the temperature, humidity and wind speed and inputting the current speed at a point 115 meters before the intersection, the system can calculate the maximum and minimum recommended speed range to provide the driver with the appropriate speed recommendation.

The system takes the input of environmental parameters and current speed in real time, takes these factors into account, and applies the appropriate algorithms and rules to calculate the exact speed recommendation range. This recommended range is dynamically adjusted according to the actual conditions of temperature, humidity, wind speed and current vehicle speed in the first 115 meters of the intersection. Ensuring safety and efficiency through intersections is critical, and an accurate speed recommendation range allows drivers to choose the appropriate speed to drive while maintaining safety to better suit the current environmental conditions of the intersection.

VI. CONCLUSION

Synthesizing the above research, this paper proposes a multi-factor speed recommendation model based on assisted driving system, which considers several key factors, such as weather conditions, traffic flow and vehicle type. By reasonably weighing the combined effects of these factors, it can provide accurate speed recommendations for driving vehicles to cross intersections quickly and smoothly. Through simulation experiments, this paper uses a Java back-end development project to verify the effectiveness of this vehicle speed recommendation model. The experimental results show that the model can provide reliable speed recommendation solutions for vehicles with assisted driving systems in different intersection scenarios. It is worth noting that the model has the adaptive capability to flexibly adjust the speed recommendation according to different road and traffic conditions. However, further research is needed to connect the model with real-time traffic data and back-end databases to verify its feasibility and reliability in real-world traffic environments.

Future research directions will focus on the measurement of real-time data and the connection of back-end engineering to the actual database and will also include further optimization and improvement of the model to improve the accuracy and intelligence of speed recommendations. In

addition, we can consider introducing more factors, such as road state and pedestrian behaviour, to further improve the comprehensive performance of the model. The future goal is to apply this speed suggestion model to a real-world assisted driving system and apply it to a real traffic environment to improve the safety, efficiency and reliability of intersection assisted driving technology and contribute to the development of future intelligent assisted driving systems.

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